DETERMINANTS OF WAGE INEQUALITY IN THE UNITED STATES

AN EMPRICAL ECONOMETRIC ANALYSIS USING 2023 IPUMS ACS MICRODATA

2025

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Roles

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  - Modeling Alternatives, Results Interpretation, Conclusion
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  - Abstract, Introduction, References

Abstract

This study analyses how the difference in education level affects income distribution in the U.S. economy. Drawing from both theoretical and empirical insight, this study explores whether an increase in education influences wage disparity or unequal levels in education deepen the existing gap between income inequality.

The paper examines three main channels through which education affects income inequality. This includes the premium effect, where people with higher education significantly increase their productivity and wages; the opportunity access effect, which determines who accesses quality education; and the structure of economic effectiveness, where technology changes how the labor market interacts with the demand and supply of educated workers.

While results from other studies indicate that higher education improves individual earning potential, unequal access to quality education contributes to widening the income gap of becoming rich or poor, particularly between highly skilled and low-skilled labor. The findings from these studies suggest that promoting equal access to education, improving the quality of education and strengthening vocational and technical pathways are crucial for reducing long-term income inequality.

Generally, the study highlights education as an important pathway to upward mobility and structural factors that can either reduce or reinforce inequality depending on the opportunities available.

1. Introduction

For several decades, income inequality has been one of the most persistent, enduring challenges in the 21st century. In both advanced and developing economies, the gap between high- and low-income earners continues to increase, widening concerns about social mobility, political stability and long-term economic growth within the nation (Piketty, 2014; OECD, 2020).

To resolve this problem, policymakers and researchers seek to understand the drivers of these inequalities; education emerges as one of the crucial and nuanced factors in influencing the distribution of income within society. Education is highly recognized as the number one determinant of individual productivity, skill acquisition and earning potential (Becker 1993). Yet on the other hand, the relationship between education and inequality goes far beyond personal benefits; it has a major effect on the labor market, technological adaptation and intergenerational mobility. This affects the broader structure of economic opportunities available. (Goldin & Katz, 2008)

The level of education an individual has influences income inequality through multiple connected channels. It is very clear that higher educational attainment generally leads to better employment offers, creating access to high-quality skilled jobs and, in return, generating higher wages. This reduces the inequality between individuals with similar opportunities. (Card 1999)

At the societal level, unequal access to quality education can widen the level of income disparity. When the education system allows wealthy individuals to access high-quality schools while those with fewer resources rely on under-resourced institutions, the difference in the level of educational attainment can lead to structural inequality in the labor market (World Bank, 2018). This creates a conscious cycle where those born with privilege expand their boundaries, while individuals from low-income backgrounds face the barriers of upward mobility (Chetty et all, 2014)

Education has a significant impact on income inequality through multiple interconnected channels. At the basic level, individuals with higher levels of education secure better employment offers and access to high-skilled jobs and higher wages, thereby reducing the level of inequality among people with similar opportunities. (Card, 1999). At the macro level, unequal access to quality education can reinforce and widen the income disparity.

When the education system is segmented, where the rich and wealthy access quality education and the poor depend on the under-resourced institutions, this difference in education qualification leads to structural inequality in the labor market (World Bank, 2018)

This eventually creates a cycle where those born into privileged families maintain or increase their advantage while individuals from low-income families or backgrounds face challenges to their upward mobility (Chetty et al, 2014)

Aside from being born into a privileged family, recent global trends have further complicated this relationship. The changes in technology and globalization have increased the demand for highly skilled labor, thereby widening the wage gap between skilled and unskilled workers (Autor, 2014). The difference in these skill-biased technological changes implies that more education is always valuable in the labor market, while increasing the income disparity gap between a worker with higher education and those with minimal education. (Acemoglue & Autor, 2011). While this is true, many economies are experiencing rapid growth in tertiary education without a corresponding match in job opportunities, thereby contributing to an oversupply of graduates in some fields and underemployment in other industries. (Altbach, Reisberg, & Rumbley, 2009). This developmental trend also challenges the traditional assumption that more education automatically leads to lower inequality.

The effect of education on income disparity can be neither simple nor uniform; it reveals the interaction between access to education, labor market dynamics and the broader social economic structure. In a society where education is made equitable, funded and aligned with labor market

needs, increasing levels of education tends to reduce inequality by improving opportunities for the disadvantaged elite (UNESCO,2021). Conversely, in economies where access to education is uneven and labor market primarily rewards individuals with specialized skills, education can become a major driver of income disparity. (Breen & Jonsson, 2005)

Given these complexities, there is a need to adopt a multidimensional approach to understand the effect of education on income inequalities. This study explores the mechanism through which education influences income distribution while highlighting not only economic return to education but also external factors such as institutional structures, policy development and contextual factors that impact who gains from quality education. By understanding these dynamics, the study addresses the current controversies on how our society can build good educational systems that can foster inclusive development and reduce income inequality (ILO,2020, Atkinson, 2015)

2. Data Exploration

The data used in this project comes from the 2023 American Community Survey (ACS) 1 year sample, a nationally representative survey conducted annually by the U.S. Census Bureau. The ACS provides detailed microdata on individuals demographics, employment, and income for the U.S. population. This dataset is cross-sectional, capturing individuals at a single point in time rather than tracking the same respondents over multiple periods.

The key variables used in this analysis include:

- **INCWAGE**: Annual wage and salary income in U.S. dollars. This is the dependent variable in the regression model. Since wage income is strongly right skewed and contains outliers, log transformation is applied for visualization and modeling.
- **EDUC**: Highest level of educational attainment (coded 0-11). Categories range from no schooling (0) to five or more years of college (11). Education is a primary explanatory variable due to its strong theoretical connection with earnings.
- **AGE**: Respondent's age in years. Age is included because labor market earnings typically follow a concave lifecycle pattern, increase early in careers and leveling off at older ages.
- **SEX**: Gender (1 = male, 2 = female). Used to capture gender differences in earnings.
- **RACE:** Detailed race categories (1-9), ranging from White (1) to multiracial classifications. This variable allows assessment of wage differences across racial groups.
- **METRO**: Metropolitan status (0 = indeterminable, 1 = non-metro, 2-4 = metro categories). Codes 2-4 reflect whether an individual lives in a metropolitan area, an important determinant of wage differences.

- **STATEFIP**: State of residence (51 unique codes, representing the 50 states plus the District of Columbia). This variable is included to control for geographic variation in labor market conditions.
- **OCC**: Occupation code (detailed, hundreds of categories).

After cleaning, the sample contains 1,658,035 observations.

2.1 Exploratory Data Analysis

Variable	Mean	Std. Dev.	Min	Median	Max
INCWAGE	63,953.82	79,051.06	4	45,000	870,000
EDUC	7.84	2.37	0	7	11
AGE	42.98	15.61	16	42	96
SEX	1.48	0.50	1	1	2
RACE	2.54	2.60	1	1	9
STATEFIP	27.95	16.09	1	28	56
OCC	3999.35	2720.56	10	4020	9830
METRO	2.64	1.40	0	3	4

Table 1: Summary Statistics

Wage income exhibits a very large range (from $4 to $870,000) and a standard deviation that exceeds the mean, confirming substantial skewness and the presence of high-income outliers. EDUC has an average value near 8, corresponding roughly to "some college", with the full range of education levels represented. AGE ranges from 16 to 96, with a mean of 43, consistent with a broad working age population. The categorical variables for SEX, RACE, METRO, and STATEFIP all display appropriate diversity, and their summary statistics reflect the distribution of coding values rather than meaningful numerical magnitudes. OCC has wide dispersion because it contains hundreds of occupation codes.

Figure 1: INCWAGE Histogram

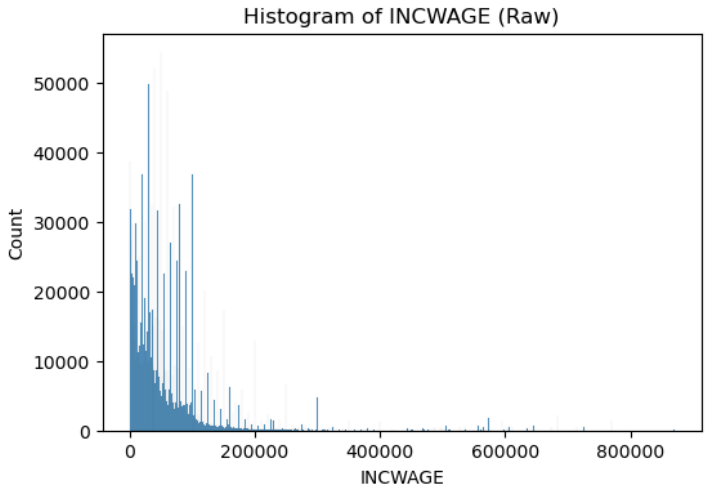
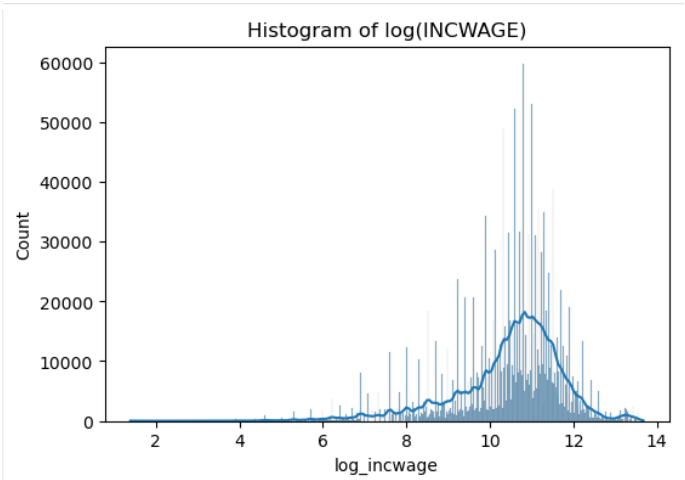


Figure 2: log(INCWAGE) Histogram



The histogram of raw INCWAGE (Fig 1) shows substantial right skewness, with most individuals earning below $80,000 while a small number earns several hundred thousand dollars. Because this skew compresses the lower end of the distribution, the income variable is transformed into log(INCWAGE). The histogram (Fig 2) and kernel density estimate of log(INCWAGE) reveal a much smoother, approximately unimodal distribution, which better meets regression assumptions and improves interpretability.

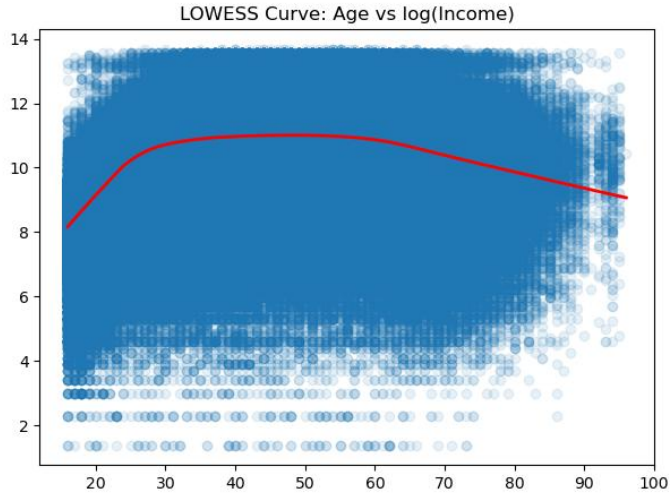


Figure 3: Scatterplot of Age vs log(INCWAGE) with LOWESS Curve

This scatterplot of AGE versus log(INCWAGE) (Fig 3) reveals the classic concave lifecycle earnings profile. Income quickly rises during early adulthood, peaks in mid-career, and gradually declines at older ages. A LOWESS curve was overlaid on the scatterplot to emphasize the curvature. This visual evidence strongly supports including both AGE and AGE^2 in the regression model.

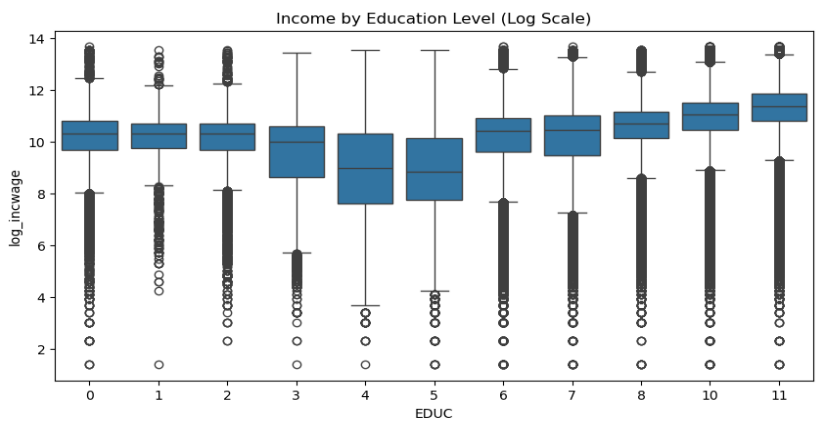


Figure 4: Boxplot of log(INCWAGE) by Education Levels

The boxplot of log(INCWAGE) across the 12 EDUC categories (Fig 4) show a clear, monotonic relationship between education levels and earnings. Median income rises consistently with each increase in educational attainment, and the interquartile range narrows among higher educated individuals. These findings align with expectations and strongly motivate including education as a categorical regressor.

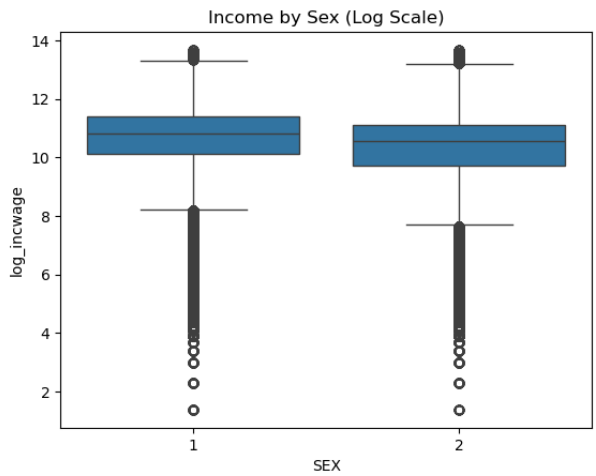


Figure 5: Boxplot of log(INCWAGE) by Sex

The boxplot for SEX (Fig 5) shows higher median income for males than for females, consistent with documented gender wage gap disparities. The difference appears both in the median earnings and dispersion, suggesting systematic differences in wage outcomes.

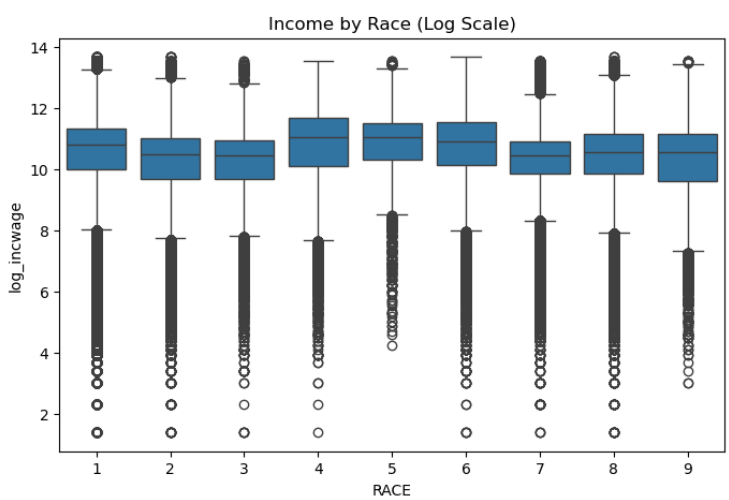


Figure 6: Boxplot of log(INCWAGE) by Race

The boxplot across the nine RACE categories (Fig 6) reveal clear differences in earnings distributions among racial groups. Several groups show lower medians or greater dispersions, while the Chinese, Japanese, and Other Asian categories (4, 5, and 6 respectively) display notably higher median log incomes compared to other racial groups. These patterns indicate persistent racial wage differentials within the dataset.

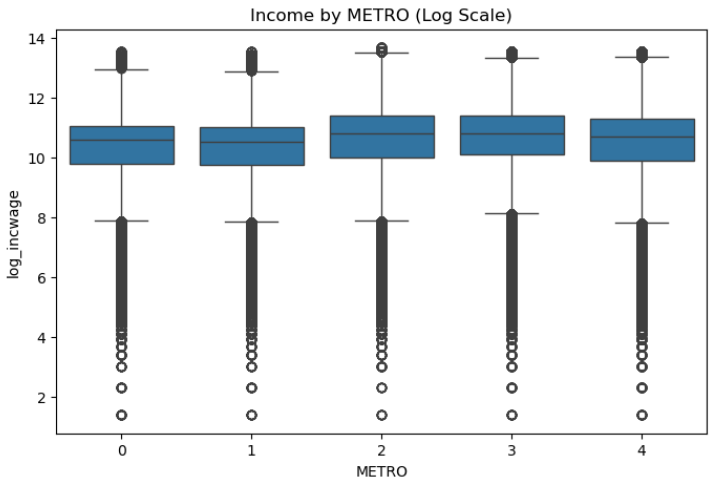
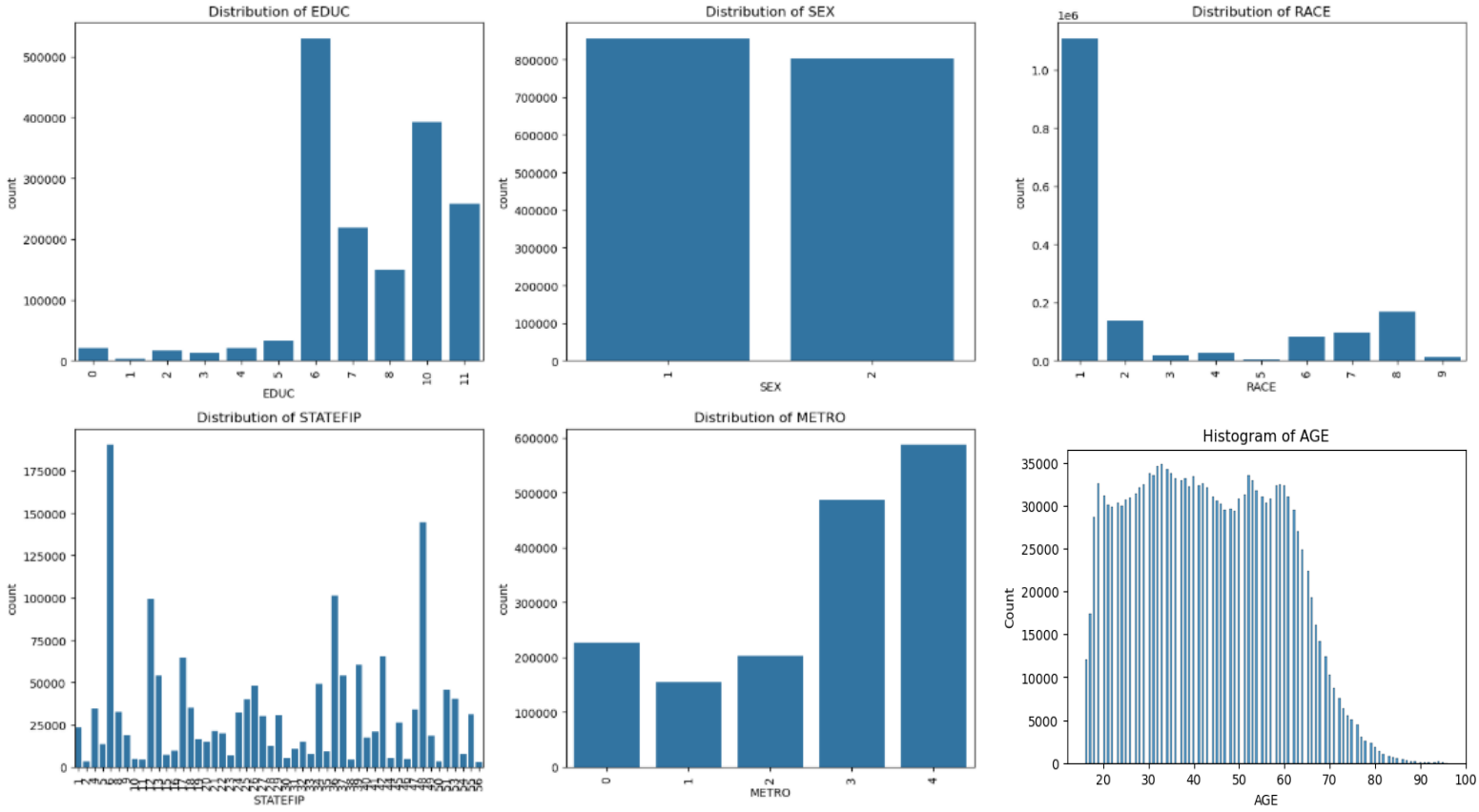


Figure 7: Boxplot of log(INCWAGE) by Metropolitan Status

The boxplot for METRO categories (Fig 7) indicate noticeable differences in the upper tail of the wage distribution and median earnings. While the lower tails appear similar across metropolitan and non-metropolitan groups, individuals residing in the metropolitan areas (codes 2, 3, and 4) display substantially higher upper-tail earnings relative to non-metro respondents. This suggests that metro areas contain a greater share of high-wage jobs.



Additional visualizations include:

- A histogram of AGE showing a broad working-age distribution (bottom right).
- Bar charts of EDUC, SEX, RACE, STATEFIP, and METRO showing frequency of each category.

These visualizations highlight the demographic and geographic diversity of the sample, illustrating variation in age, education, racial composition, gender, metro status, and state of residence that supports meaningful wage comparisons in the regression analysis.

The exploratory analysis provides several insights that guide the upcoming model specification:

1. **Income is highly skewed,** justifying the use of log(INCWAGE) in regression.
2. **Age shows a strong concave relationship with income**, motivating the inclusion of AGE^2.
3. **Education, sex, race, and metropolitan status exhibit clear earnings differences**, indicating strong predictive power.
4. **State and occupation categories vary widely**, supporting their inclusion as fixed effects rather than numeric variables.

These patterns provide a strong foundation for the regression modeling presented in the next section.

3. Model & Results

In this section, we compare different models we experimented with, but we will focus primarily on our best-performing model.

As we saw in the EDA section, INCWAGE has a strong right skew, so we use its log transformation as our dependent variable.

Most of our predictors are categorical, so we create dummy variables with binary encoding to indicate whether each category is present. This applies to EDUC, SEX, RACE, OCC, STATEFIP, and METRO. OCC and STATEFIP have many more categories compared to the other predictors (together, they add over 650 dummy variables to the model), so we will exclude their coefficients from our analysis since their main purpose is to serve as controls for estimates of INCWAGE.

AGE is our only numerical predictor. From the EDA, we found that AGE has a strong concave relationship with INCWAGE, so we also include AGE^2 in our regression.

Finally, we include an interaction term, $EDUC \times SEX$, allowing us to see whether the impact of education on income is different for men and women.

3.1 OLS Assumptions

Before we interpret our model results, we address each OLS assumption to support the claim that our estimates are unbiased:

1. Linearity in Parameters
   - We checked for nonlinearities in our EDA and applied transformations when needed (e.g., log(INCWAGE), AGE^2).

2. Random Sampling
   - Our data was gathered from IPUMS 2023 survey data, which we consider to be a reliable source for acquiring randomly sampled data. The data we retained for our analysis contains 1.5+ million records, which increases the likelihood for it to be representative of the larger working population.

3. Multicollinearity
   - We checked for multicollinearity by calculating the Variance Inflation Factor (VIF) for each predictor variable. Since none of the VIF scores are greater than 5, we can safely assume multicollinearity will not significantly influence regression results.

Variable	**VIF**
EDUC	1.271506
AGE	1.019779
SEX	1.0239
RACE	1.050382
STATEFIP	1.018861
OCC	1.268757
METRO	1.036527

Table 2: VIFs for Each Predictor

4. Zero Conditional Mean
   - We believe our chosen variables capture most effects playing into income. There is a chance that some unmeasured effects, such as family wealth and individual motivation, contribute to omitted variable bias. However, independently collecting this information would not be possible, so we assume the impact of any excluded confounding factors is minimal.

5. Homoskedasticity
   - We used the Breusch–Pagan test to check for heteroskedasticity. The test yielded a p-value of ~0.00, so we reject the null hypothesis and conclude that heteroskedasticity is present. To correct our standard errors, we include the argument *cov_type = HC1* when fitting our model, which provides us with heteroskedasticity-robust estimates.

3.2 Model Variations

Below is a summary of the different models we tested.

Model	Major Differences	Adjusted R^2
1	• STATEFIP and OCC removed for simplicity	0.330
2	• EDUC converted to numerical EDUC_YRS • STATEFIP simplified to 4 categories (Northeast, Southeast, Midwest, West) • OCC simplified to 5 major occupation categories: - Management, Business, Science, and Arts - Service - Sales and Office - Natural Resources, Construction - Maintenance, Production, Transportation, and Material Moving • METRO includes 3 categories (Non-metro, Indeterminate, Metro)	0.345
3	• All categories for STATEFIP and OCC included (650+) • METRO includes 2 categories (Non-metro, Metro)	0.439

Table 3: Comparison of Model Alternatives

Below is the regression output of our best-performing model (with estimates for STATEFIP and OCC omitted)**.**

3.3 Best Model & Interpretation

We ran this model with the following equation:

$$\log(INCWAGE_i) = \beta_0 + \beta_1 AGE_i + \beta_2 AGE^2{}_i + \beta_3 EDUC_i + \beta_4 SEX_i + \beta_5 RACE_i + \beta_6 METRO_i + \beta_7(SEX_i * EDUC_i) + \gamma_s + \delta_o + u_i$$

with γ_s and δ_o representing state and occupation fixed effects.

Variable	(1)	Confidence Interval
Intercept	8.2285*** (0.017)	(8.196, 8.261)
AGE	0.1369*** (0.0004)	(0.136, 0.138)
AGE^2	-0.0014*** (0.00000424)	(-0.001, -0.001)
Education (baseline = no schooling)		
Nursey School to Grade 4	0.0358 (0.022)	(-0.007, 0.079)
Grade 5, 6, 7, or 8	0.0564*** (0.013)	(0.030, 0.082)
Grade 9	-0.1882*** (0.016)	(-0.220, -0.156)
Grade 10	-0.3794*** (0.015)	(-0.408, -0.350)
Grade 11	-0.3825*** (0.013)	(-0.409, -0.356)
Grade 12	0.1314*** (0.010)	(0.112, 0.151)
1 year of college	0.1063*** (0.010)	(0.086, 0.126)
2 years of college	0.2185*** (0.010)	(0.198, 0.239)
4 years of college	0.3954*** (0.010)	(0.375, 0.415)
5+ years of college	0.5500*** (0.010)	(0.530, 0.570)
Race (baseline = White)		
Black/African American	-0.1084*** (0.003)	(-0.114, -0.102)
American Indian	-0.0664*** (0.008)	(-0.082, -0.051)
Chinese	-0.0636*** (0.006)	(-0.076, -0.052)

Japanese	-0.0694*** (0.016)	(-0.100, -0.039)
Other Asian	-0.0843*** (0.004)	(-0.091, -0.077)
Other race	-0.0112*** (0.003)	(-0.018, -0.005)
Two major races	-0.0406*** (0.003)	(-0.046, -0.035)
Three or more major races	-0.0843*** (0.009)	(-0.102, -0.066)
SEX (baseline = Male)		
Female	-0.2266*** (0.016)	(-0.258, -0.196)
Metro Area (baseline = non-metro)		
In Metro	0.08866*** (0.002)	(0.085, 0.093)
EDUCATION x FEMALE Interaction Term		
Nursey School to Grade 4	0.0387 (0.036)	(-0.032, 0.109)
Grade 5, 6, 7, or 8	-0.0804*** (0.022)	(-0.124, -0.037)
Grade 9	-0.0382 (0.026)	(-0.090, 0.013)
Grade 10	0.0023 (0.023)	(-0.043, 0.047)
Grade 11	0.0522** (0.021)	(0.012, 0.093)
Grade 12	-0.0198 (0.016)	(-0.051, 0.012)
1 year of college	-0.0164 (0.016)	(-0.048, 0.016)
2 years of college	-0.0499*** (0.016)	(-0.082, -0.018)
4 years of college	-0.0264 (0.016)	(-0.058, 0.005)

5+ years of college	-0.0096 (0.016)	(-0.041, 0.022)

	Model Statistics
Number of Observations	1,658,035
R-squared	0.439
Adj R-squared	0.439
Robust SE	Yes
Method	OLS (HC1 robust)

Notes: Robust standard errors in parentheses below respective coefficients. Model includes age and age^2. Fixed effects (Occupation and State) are not included in the table. *** p < 0.01, ** p < 0.05, * p < 0.10.

Our preferred model has the highest adjusted R-squared value of 0.439, meaning the predictors included explain nearly 44% of the variability in income. In contrast, the alternative models, which simplify or omit STATEFIP and OCC, have adjusted R-squared values roughly 0.10 lower. This indicated to us that including these variables in full was important to maximize the explanatory power of the model.

From our log-linear model we can clearly identify the relative impact of education on income at various levels. For instance, we see a coefficient of 0.5500 for "5+ years of college," indicating that this group is associated with a $100\% * (e^{0.55} - 1) = 73.33\%$ higher income than the baseline (i.e., those with no schooling). Similarly, a high school education ("Grade 12") is associated with a ~13% increase in income compared to the baseline. Each estimate is accompanied by a confidence interval, which gives us a range of values over which we are 95% confident the true coefficient lies. For Grade 12, the coefficient 0.131 lies in the middle of the confidence interval (0.112, 0.151), meaning there is an uncertainty of ± 0.019.

To be sure our coefficients are statistically significant, we would need to perform a hypothesis test. For the "Grade 12" dummy variable, we start with $H_0: \beta_{Grade12} = 0 \; and \; H_1: \beta_{Grade12} \neq 0$. We calculate our test statistic from the parameter estimate and standard error: $t = \frac{\widehat{\beta_{Grade12}}}{SE(\widehat{\beta_{Grade12}})} = \frac{0.1314}{0.010} = 13.14$. Given $df = n - k = 1{,}658{,}035 - 612 = 1{,}657{,}423$ degrees of freedom, our critical value at the 5% level is 1.96. Since $|t| > |t_{critical}| = 13.14 > 1.96$, we reject H_0 and conclude that the coefficient is statistically significant. In the regression table, we have used asterisks to label each estimate as being significantly different from zero at the 10% (*), 5% (**), or 1% (***) levels.

Considering our model results as a whole, we can draw several broader conclusions. EDUC, SEX, and AGE have the largest impact on relative change in income. Higher education, especially beyond 2 years of college, is associated with a significant (+20%) increase in income. Conversely, being a female worker is associated with a 20% decrease in income, reinforcing existing evidence of a gender wage gap. For age, the marginal effect on income is given by $\beta_{AGE} + 2\beta_{AGE2} * AGE$. For someone who is 25 years old, an additional year of age is associated with a $0.1369 - 2(0.0014)(25) = 0.0669 \rightarrow 6.69\%$ increase in income, whereas someone who is 45 is only expected to have a $0.1369 - 2(0.0014)(45) = 0.0109 \rightarrow 1.09\%$ increase in

income. These diminishing returns continue until approximately age 49, where an additional year is associated with a negative effect on income. The negative impact on income then grows with each year beyond this point. Finally, our interaction term, $EDUC \times SEX$, has almost no statistically significant coefficients, so we cannot definitively say that the effect of education on income differs between men and women.

For this model, the total sum of squares (SST) is 2,791,173.63, reflecting substantial variability in log wages across individuals. The explained sum of squares (SSR) is 1,225,316.43, indicating that the included regressors account for a large share of variation in earnings. The residual sum of squares (SSE) is 1,565,857.20, capturing unexplained variation that is attributed to unobserved factors. The relationship SST = SSR + SSE holds, confirming the variance decomposition. Using both SSR and SST, we can calculate the R-squared with the following formula: $\frac{SSR}{SST} = \frac{1{,}225{,}316.43}{2{,}791{,}173.63} \approx 0.439$. We can see that calculated value matches the reported R-squared earlier, which is relatively high for cross sectional wage regressions.

3.4 Limitations

This study has several key limitations. First, it focuses on a snapshot of data in 2023. Any inferences made in this study apply only to this point in time, as we are unable to factor in changing conditions over time. Additionally, while we aimed to build a comprehensive regression model, we acknowledge there will be some degree of omitted variable bias from factors we cannot observe, such as family wealth and individual motivation. Finally, this study focuses on income from wages, which is the main source of income for most Americans, but it excludes other sources such as passive or investment income. This limits the model's ability to capture the true level of economic inequality.

4. Conclusion

The goal of this study was to understand how educational attainment influences income inequality in the U.S. Using 2023 IPUMS ACS microdata, we developed an econometric model with various demographic and geographic controls. We found that educational attainment at or beyond 2 years of college is associated with a significant increase in income, while having less than 12 years of schooling is associated with a modest or even negative change in income compared to those with no schooling. Among our controls, age and sex were the most significant, with additional age showing a positive impact until 49, and sex showing an associated 20% gap in income between men and women, reflecting current estimates in the gender wage gap.

From this study, we conclude that education is a significant predictor of income. Thus, government and institutions should invest further to provide more affordable education or access to specialized training for skills that are in line with current market demands. This would level

the playing field and help alleviate income inequality in the U.S. Future work should incorporate longitudinal data to analyze trends over time and account for changing economic conditions. Additionally, it should explore methods to collect or estimate supplemental income, as well as excluded predictors such as generational wealth.

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